

1.

- MODULE and SESSION
 - There is a one-to-many relationship between a Module and a Session.
 - A MODULE can have many Sessions, but a Session can only belong to one Module.
- SESSION and SCHEDULE
 - The relationship between Session and Schedule is a one-to-many relationship.
 - A Session may be Scheduled zero or many times, but each Schedule must belong to exactly one Session
- STUDENT and SCHEDULE
 - The relationship between a Student and Schedule is a one-to-many relationship.
 - A Student must have at least one Schedule, but each Schedule is associated with exactly one Student.

2. a)

Table	Primary Key	Foreign Key(s)
STUDENT	stuNum	staffNum
STAFF	staffNum	N/A

b)

Table	Yes / No	Explanation
STUDENT	Yes	Each row has a unique identity, and the stuNum attribute is never null. ✓
STAFF	Yes	Each row has a unique identity, and the staffNum attribute is never null. ✓

c)

Table	Yes / No	Explanation
STUDENT	No	The staffNum(FK) 17860 points to a nonexistent entry in the STAFF table
STAFF	No	The table doesn't have

		foreign keys
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3. a) $\Pi_{stuNum}(STUDENT)$

RESULTANT TABLE

stuNum
22405887
22365169
22478172
22233435
22692868
22345427

b) $\Pi_{stuNum, deptID}(STUDENT \bowtie STAFF)$

RESULTANT TABLE

stuNum	deptID
22365169	2119
22478172	1871
22233435	2119
22345427	2034

4. a)

- A Unary relationship exists when an entity is related to itself. An example in this case would be a SalesPerson managing other SalesPersons. In this relationship, one SalesPerson plays the role of a Manager, and the other one plays that of a Subordinate. ✓
- A Binary relationship is an association between two entities. This is depicted by the relationship between Product and ProductCategory. ✓
- A Ternary relationship is a relationship between three entities. An example in this instance is a SalesPerson sells a Product to a Customer. This is a ternary relationship because the associative entity Sale represents a single transaction event that ✓

simultaneously involves all three parent entities (SalesPerson, Product and Customer).

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b) SALE is a weak entity. The existence of a Sale depends on the Product being sold, the Customer making the purchase, the SalesPerson who assisted with the Sale, and it also has primary keys that are derived from its parent entities.

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c) If a new saleID attribute is added to SALE, it would still be a weak entity. This is because Sale is an associative entity that represents a single transaction event that simultaneously involves three parent entities. So, a Sale only makes sense when a Salesperson sells a Product to a Customer. All of these make Sale existence-dependent on SalesPerson, Product and Customer. Another reason is that saleID is added as a surrogate key, which implies that function-wise, the sale entity doesn't change. The key is added only to ease the process of implementing the database.

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Are the fks mandatory or not? Why?
What is the natural pk?